

Foundation of Data Science

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Prerequisites

- ❑ Basic set theory
- ❑ Combinatory
- ❑ Calculus (knowing derivatives and integrals)
- ❑ Probability theory
- ❑ Statistics
- ❑ Programming (MATLAB/Python)

Data

- ❖ **Data is everywhere around us**
- ❖ **Volume of data generated everyday is very high**
- ❖ **Data can be generated by**
 - ❖ **Humans**
 - ❖ **Machines or**
 - ❖ **Humans-machines combines**

Imagine processing, analyzing and managing this much of data

How to deal with this much of data?

Data Science

- **Data science** is the science of **analyzing raw data** using ***statistics*** and ***machine learning*** techniques with the purpose of drawing insights from the data
- **Data science** is used in many industries to allow them to make **better business decisions** and in the sciences to test models or theories
- This requires a process of inspecting, cleaning, transforming, modeling, analyzing and interpreting raw data



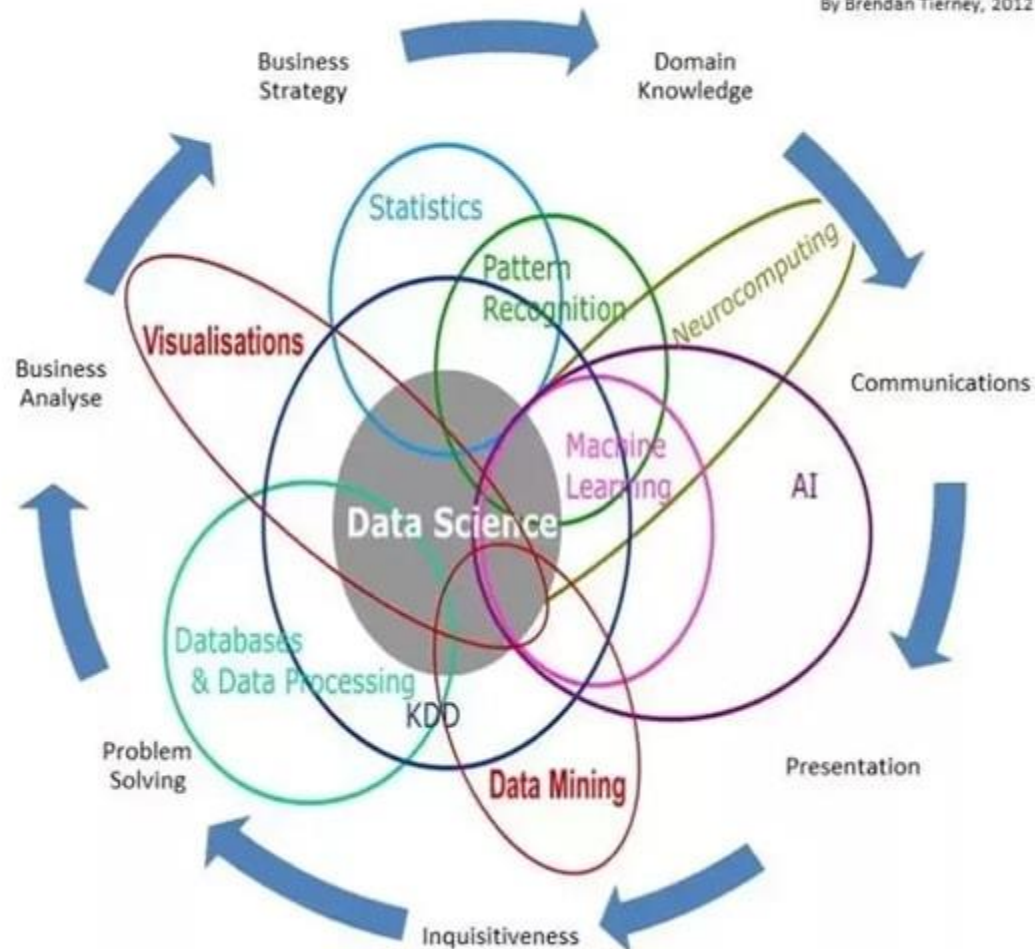
Data Science

- **Data science** “uses scientific methods, processes, algorithms and systems to extract knowledge and insights from data in various forms”
- **Data science** is an umbrella term that encompasses **data analytics, data mining, machine learning, and several other related disciplines.**
- While **a data scientist** is expected to forecast the future based on past patterns, **data analysts** extract meaningful insights from various data sources.
- A **data analyst** is usually the person who can do basic descriptive statistics, visualize data, and communicate data points for conclusions

Data Science

Data Science Is Multidisciplinary

By Brendan Tierney, 2012



Data Scientist



ML Expert



Data Perspective

- **Read data**
- **Data processing and cleaning**
- **Summarizing data**
- **Visualization**
- **Deriving insights from data (machine learning)**

Statistics vs Machine Learning

Statistics

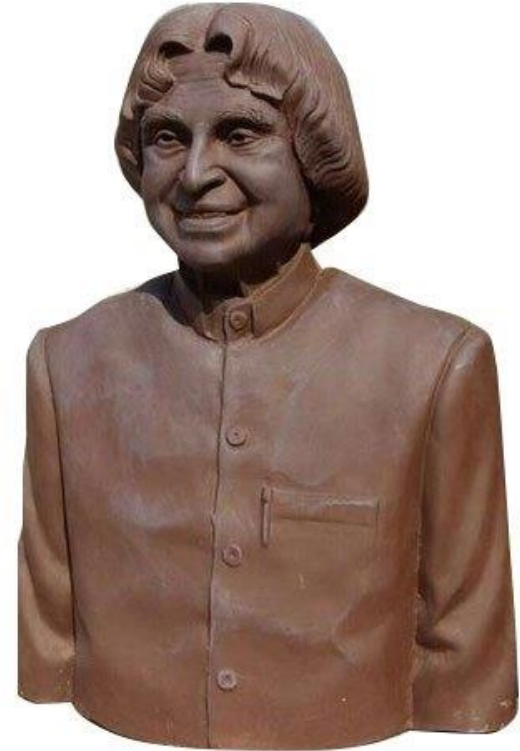
Problem -> Data

Machine Learning

Data -> Solution

Learning?

Can you recognize these images?



How do you recognize it?

Origin of Learning?

.....Lies in very early efforts of understanding Intelligence.

What is Intelligence?

- It can be defined as the ability to comprehend; to understand and profit from experience.
- Capability to acquire and Apply Knowledge.

What is Machine Learning?

- It is very hard to write programs that solves problems like recognizing handwritten documents, recognizing a 3D object from a novel viewpoint in new lighting conditions in a cluttered scene.
 - We don't know what program to write because we don't know how its done in our brain.
 - Even if we had a good idea about how to do it, the program might be extremely complicated.

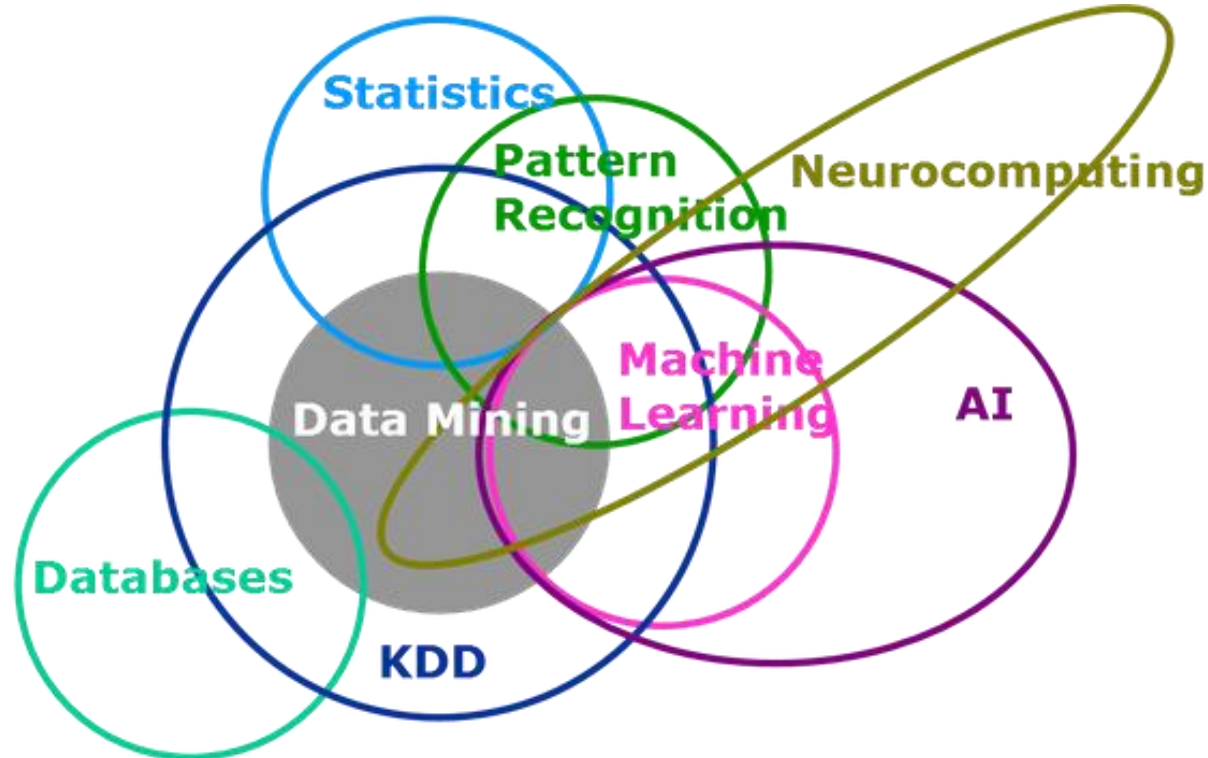
Machine Learning for Pattern Recognition

- **Learning:** Acquiring new knowledge or modifying the existing knowledge
- **Knowledge:** Familiarity with information present in data
- **Learning by machines for pattern analysis:** Acquisition of knowledge from data to discover patterns in data
- **Data-driven techniques to learning by machines:** Learning from examples (Training of models)
- **Generalization ability of learning machines:** Performance of trained models on new (test) data
- **Target of learning techniques:** Good generalization ability
- **Learning techniques:** Estimation of parameters of models
- **Learning machines and Learning techniques for pattern analysis:**
 - **Statistical Models (Maximum likelihood)**
 - **Artificial Neural Networks (Error correction learning)**
 - **Kernel Methods (Learning optimal linear relationships)**
 - **Deep Learning Models (Representation learning)**

Machine Learning

"Field of study that gives computers the ability to learn without being explicitly programmed".

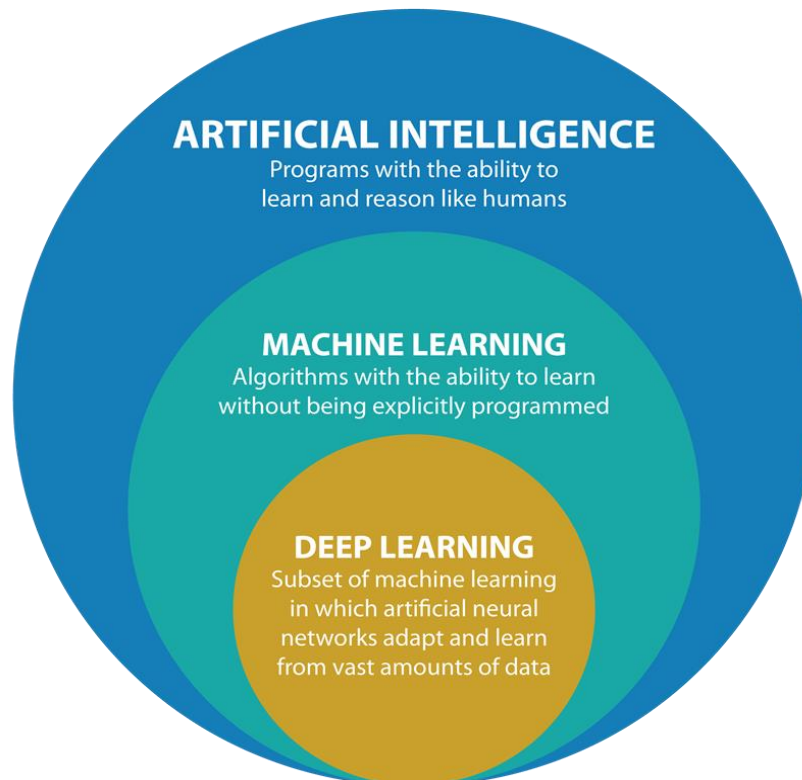
Arthur Samuel, 1959



Machine Learning

"Well posed learning problem: A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E ".

Tom Mitchell (1998)



Machine Learning

“Well posed learning problem: A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E ”.

Example:

Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. What is the task T in this setting?

- Classifying emails as spam or not spam. (Task T)
- Watching you label emails as spam or not spam. (Experience E)
- The number (or fraction) of emails correctly classified as spam/not spam. (Performance measure P)

Learning Algorithms

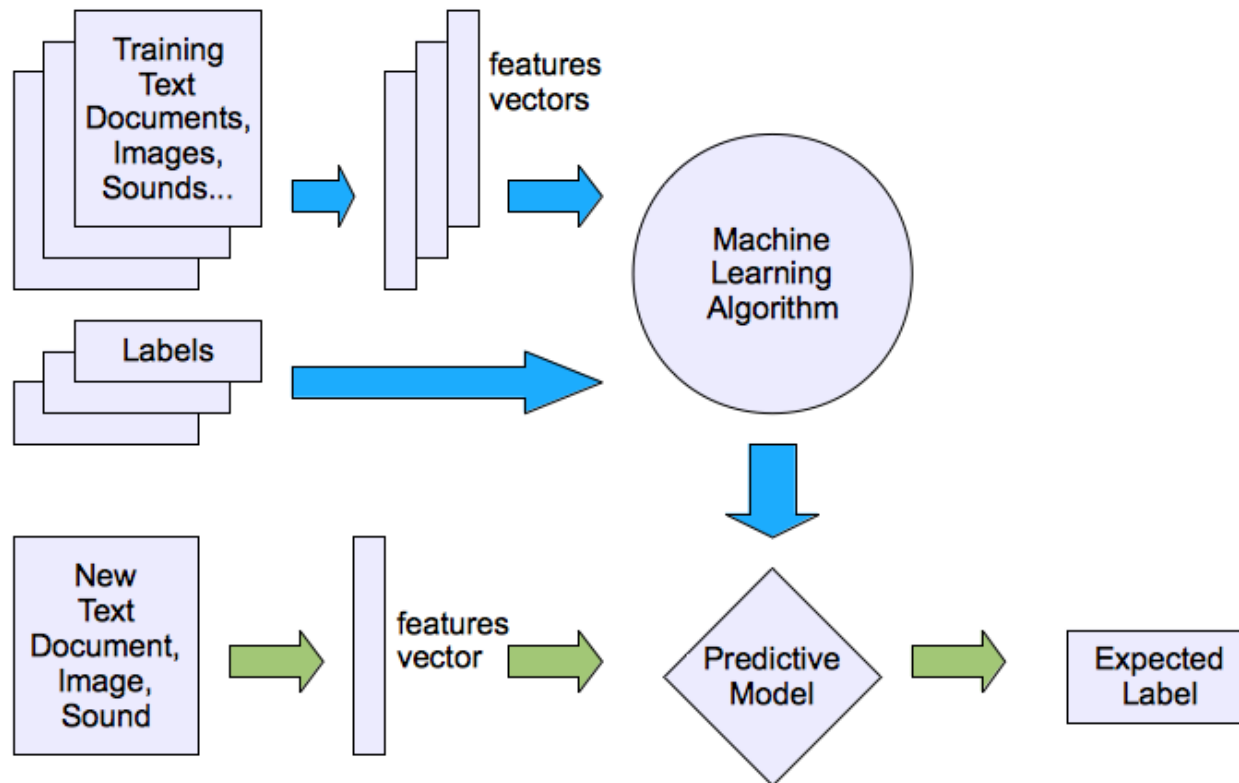
- **Supervised learning** ($\{x_n \in R^d, y_n \in R\}_{n=1}^N$)
 - Prediction
 - Classification (discrete labels), Regression (real values)
- **Unsupervised learning** ($\{x_n \in R^d\}_{n=1}^N$)
 - Clustering
 - Probability distribution estimation
 - Finding association (in features)
 - Dimension reduction (not all)
- **Semi-supervised learning**
- **Reinforcement learning**

Learning Algorithms



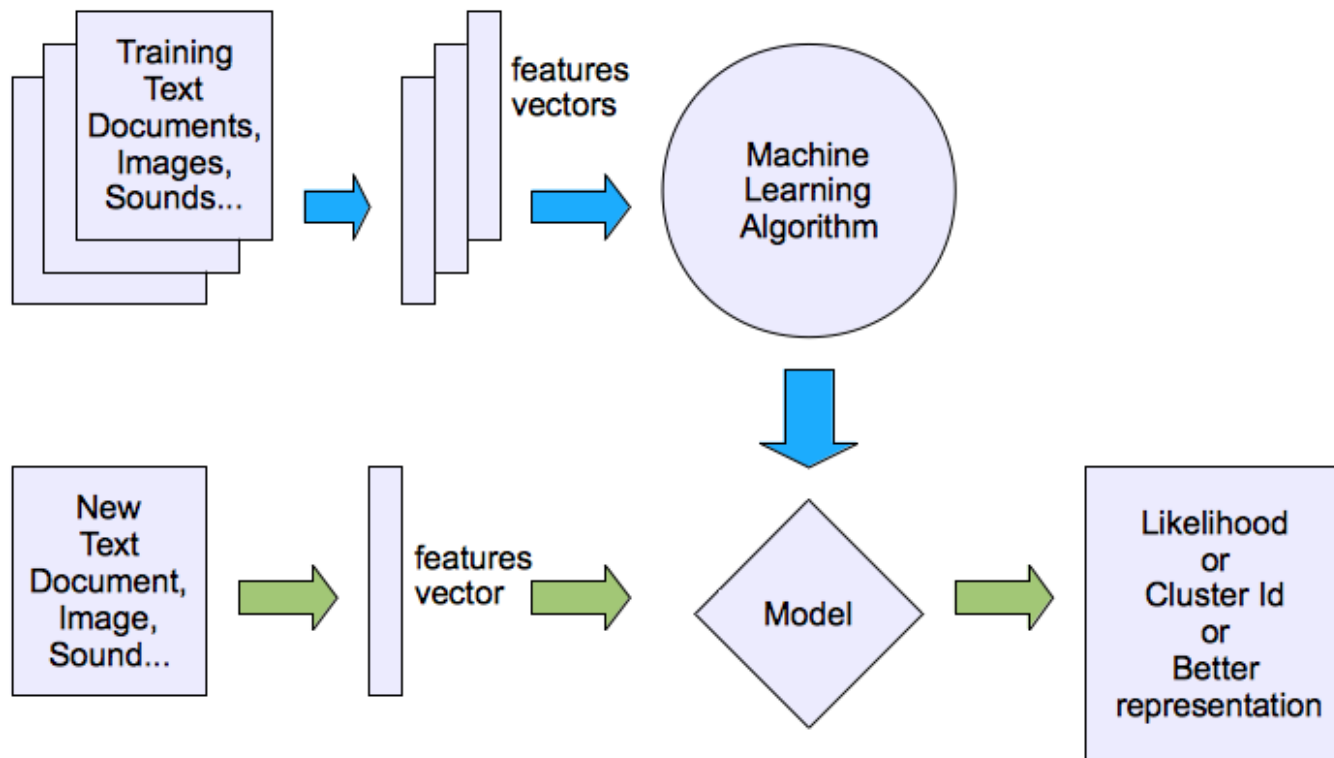
ML structure

- Supervised learning



ML structure

- Unsupervised learning



Learning Algorithms: Examples

- **Supervised learning**

- Naïve Bayes
- K-nearest Neighbor
- Support Vector Machine
- Decision Tree
- Random Forest
- Feed Forward Neural Network

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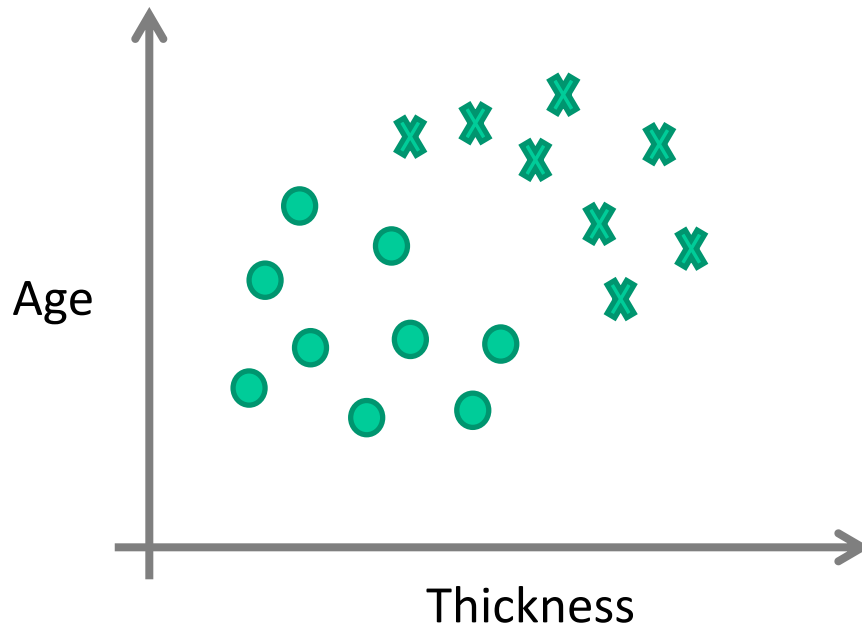
- **Unsupervised learning**

- K-means Clustering
- Principal Component Analysis

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Supervised learning example

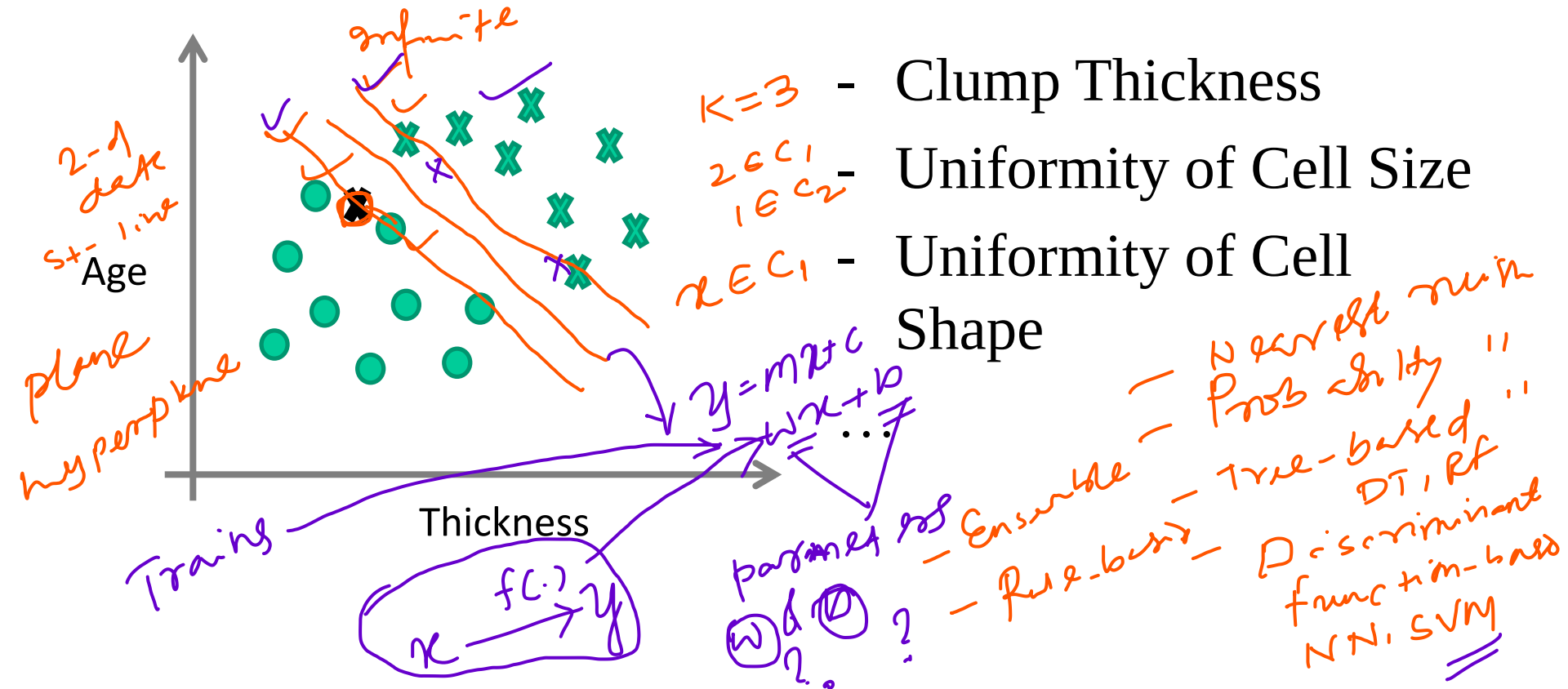
Breast cancer (malignant, benign): Number of features may be more than one



- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape
- ...

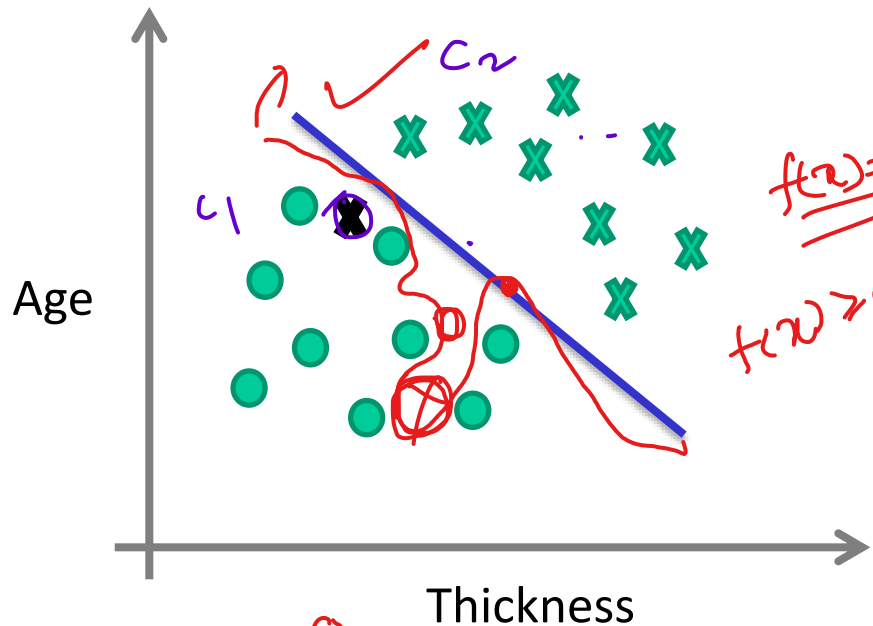
Supervised learning example

Breast cancer (malignant, benign): Number of features may be more than one



Supervised learning example

Breast cancer (malignant, benign): Number of features may be more than one



- Clump Thickness
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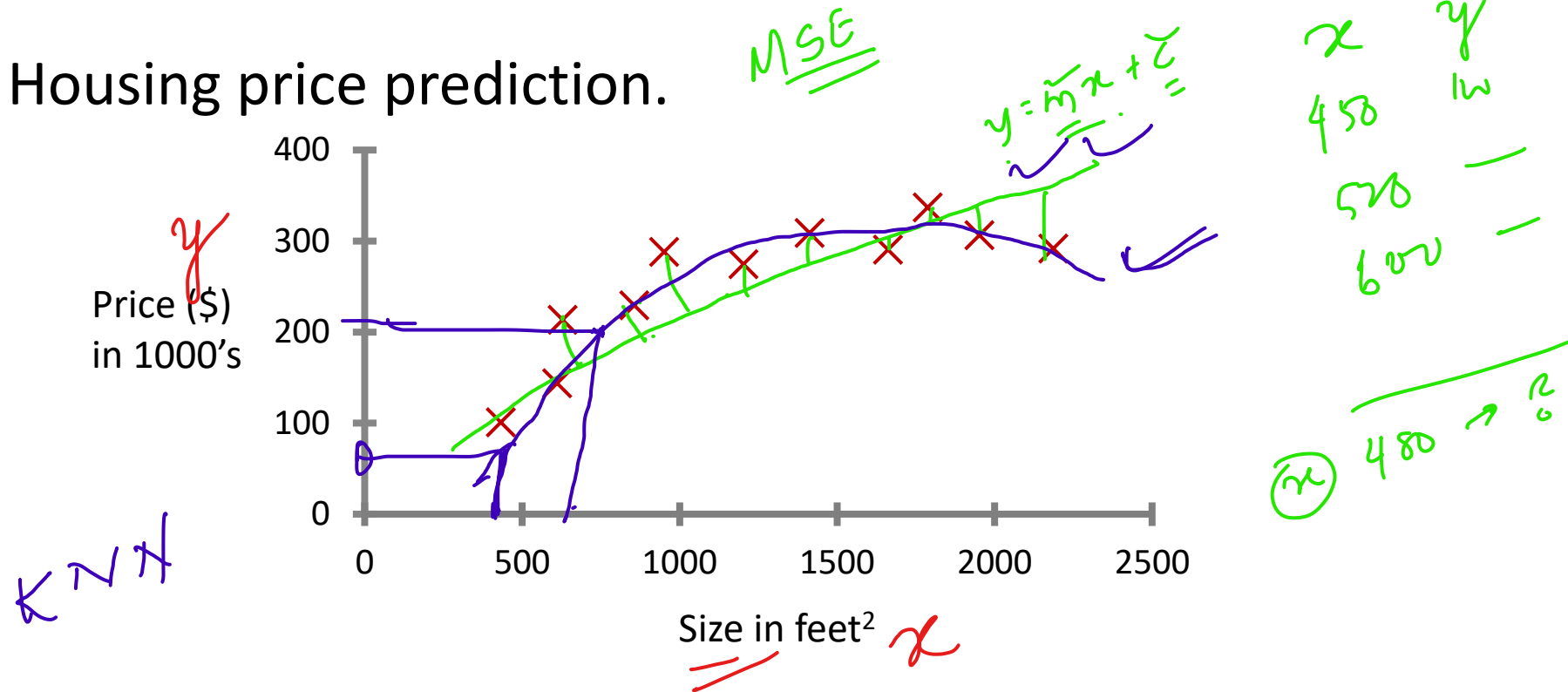
w & b

$$y = w \cdot x + b$$

$x_i \rightarrow y_i$

Supervised learning example

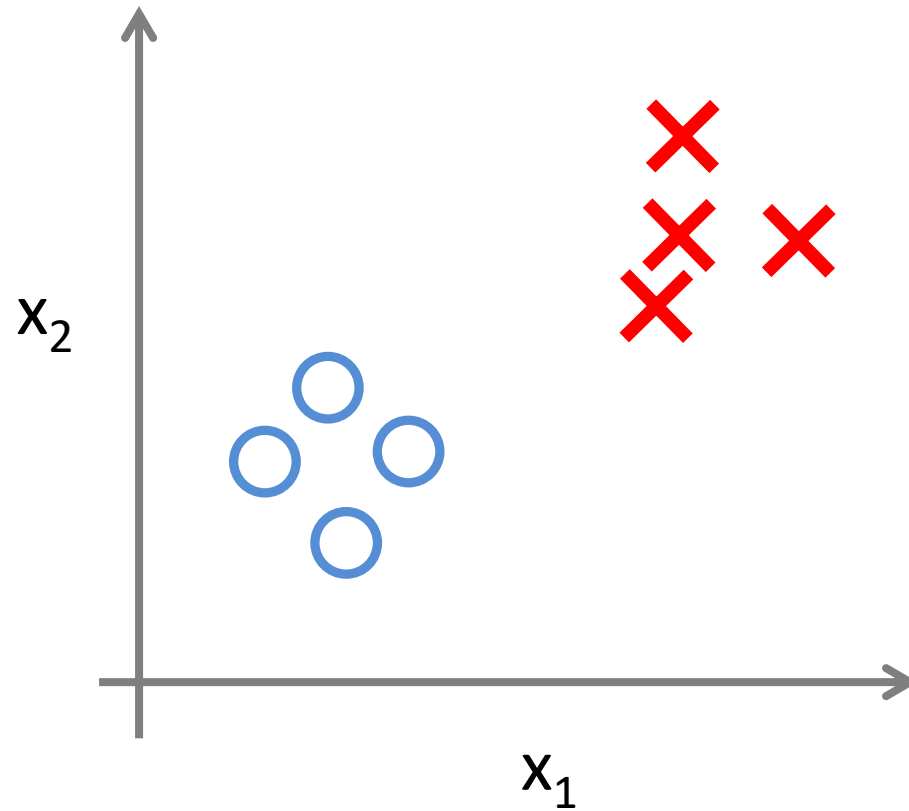
Housing price prediction.



Supervised Learning
“right answers” given

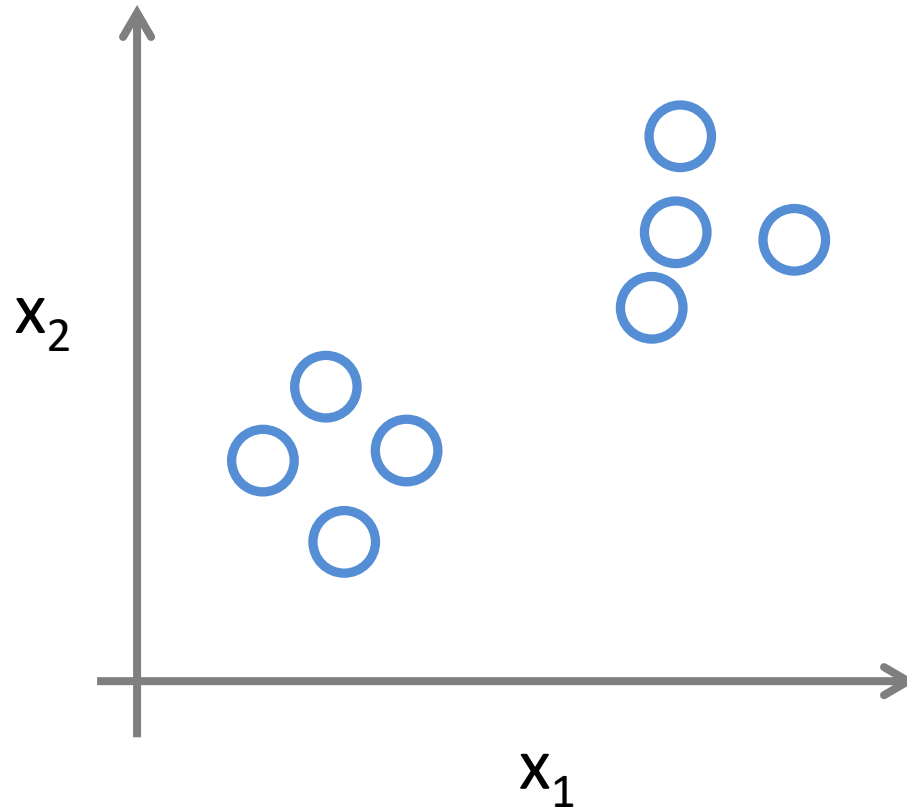
Regression: Predict continuous
valued output (price)

Supervised Learning



Unsupervised learning example

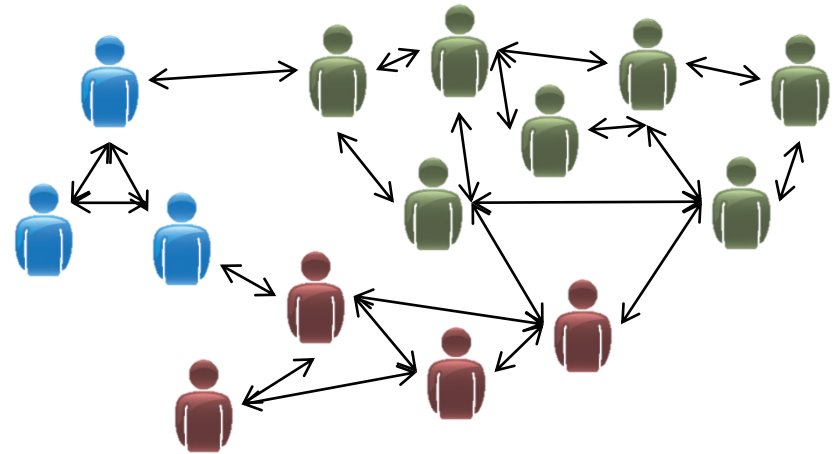
Unsupervised Learning



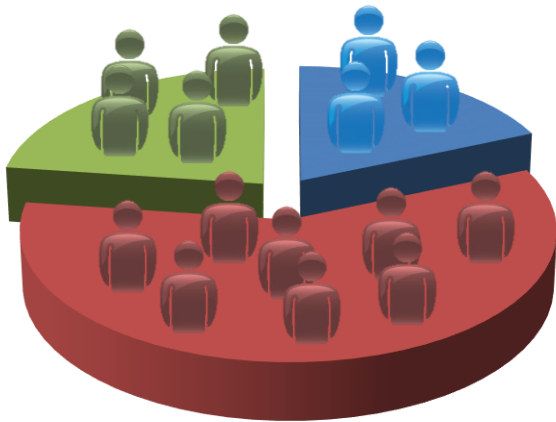
Unsupervised learning example



Organize computing clusters



Social network analysis



Market segmentation



Astronomical data analysis

Unsupervised learning example

Headlines

[More Headlines](#)

COVID-19 news: See the latest coverage of the coronavirus (COVID-19)

Pakistans new trick, claims Kulbhushan Jadhav refused to file review petition against death sentence

Zee News · 59 minutes ago



- **Kulbhushan Jadhav refused to file review petition, wants to go ahead with mercy plea, claims Pakistan**
Times of India · 48 minutes ago
- **Kulbhushan Jadhav refused to file review petition against death sentence, claims Pakistan**
TIMES NOW · 3 hours ago
- **Kulbhushan Jadhav refused to seek review of his death sentence, says Pakistan**
Hindustan Times · 6 hours ago
- **Pakistan claims Kulbhushan Jadhav refused to file review plea, MEA says he was coerced**
The Indian Express · 52 minutes ago

[View Full coverage](#)

Questions

Of the following examples, which would you address using an unsupervised learning algorithm?

- ☐ Given email labeled as spam/not spam, learn a spam filter.
- ☐ Given a set of news articles found on the web, group them into set of articles about the same story.
- ☐ Given a database of customer data, automatically discover market segments and group customers into different market segments.
- ☐ Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

Some examples of tasks best solved by machine learning

- Database mining

Large datasets from growth of automation/web.

E.g., Web click data, medical records, biology, engineering

- Applications can't program by hand.

E.g., Autonomous helicopter, handwriting recognition, most of Natural Language Processing (NLP), Computer Vision.

- Self-customizing programs

E.g., Amazon, Netflix product recommendations

- Understanding human learning (brain, real AI).

Some examples of tasks best solved by machine learning

- Recognizing patterns:
 - Object in real scenes
 - Facial identities or facial expressions
 - Handwritten documents
 - Spoken words
- Recognizing anomalies:
 - Unusual sequences of credit card transactions
 - Unusual patterns of sensor readings in a nuclear power plant
- Prediction:
 - Future stock prices or currency exchange rates
 - Which movies will a person like?
- Clustering

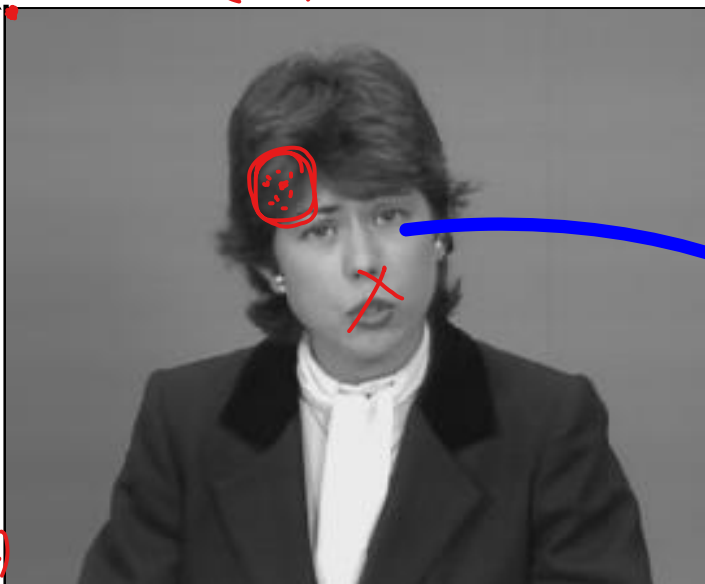
What computers see?

A Digital Image

Digital image representation

origin

$(0,0)$ $(0,1)$ $(0,2)$... $(M-1)$



$f(x,y)$
(Gray level)

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,M-1) \\ f(1,0) & f(1,1) & \dots & f(1,M-1) \\ \dots & \dots & \dots & \dots \\ f(N-1,0) & f(N-1,1) & \dots & f(N-1,M-1) \end{bmatrix}$$



- Sampling
- Quantization

$$M \times N = 2^{10} \times 2^{10} = 1024 \times 1024$$

$N \times M$ pixels

$N \times M \times 3$
 $\downarrow$
R, G, B
 $N \times M$ $N \times M$ $N \times M$

$f(x,y)$
 $(0,0) = r(0,0) \cdot g(0,0) \cdot b(0,0)$
 $(255, 255, 255)$

$f(x,y)$
 $(x,y) \rightarrow$ spatial coordinates
 $f(x,y)$

$\downarrow$
Binary
 $0 \rightarrow$ black
 $1 \rightarrow$ white
 $8\text{-bit} = 2^8 = 256$
 $\downarrow$
 0 to 255
 $\downarrow$
black to white

Feature Set

- ❑ In a classification task, we are given a pattern and the task is to classify it into one out of c classes.
- ❑ Supervised (The number of classes c is assumed to be known a priori)
- ❑ Each pattern is represented by a vector of features, $x(i), i = 1, 2, \dots, d$ which makes a d -dimensional feature vector

x^M

$$x = [x(1), x(2), \dots, x(d)]^T \in \mathbb{R}^d$$

$x_1 = [f_1 \ f_2 \ \dots \ f_{10}]$

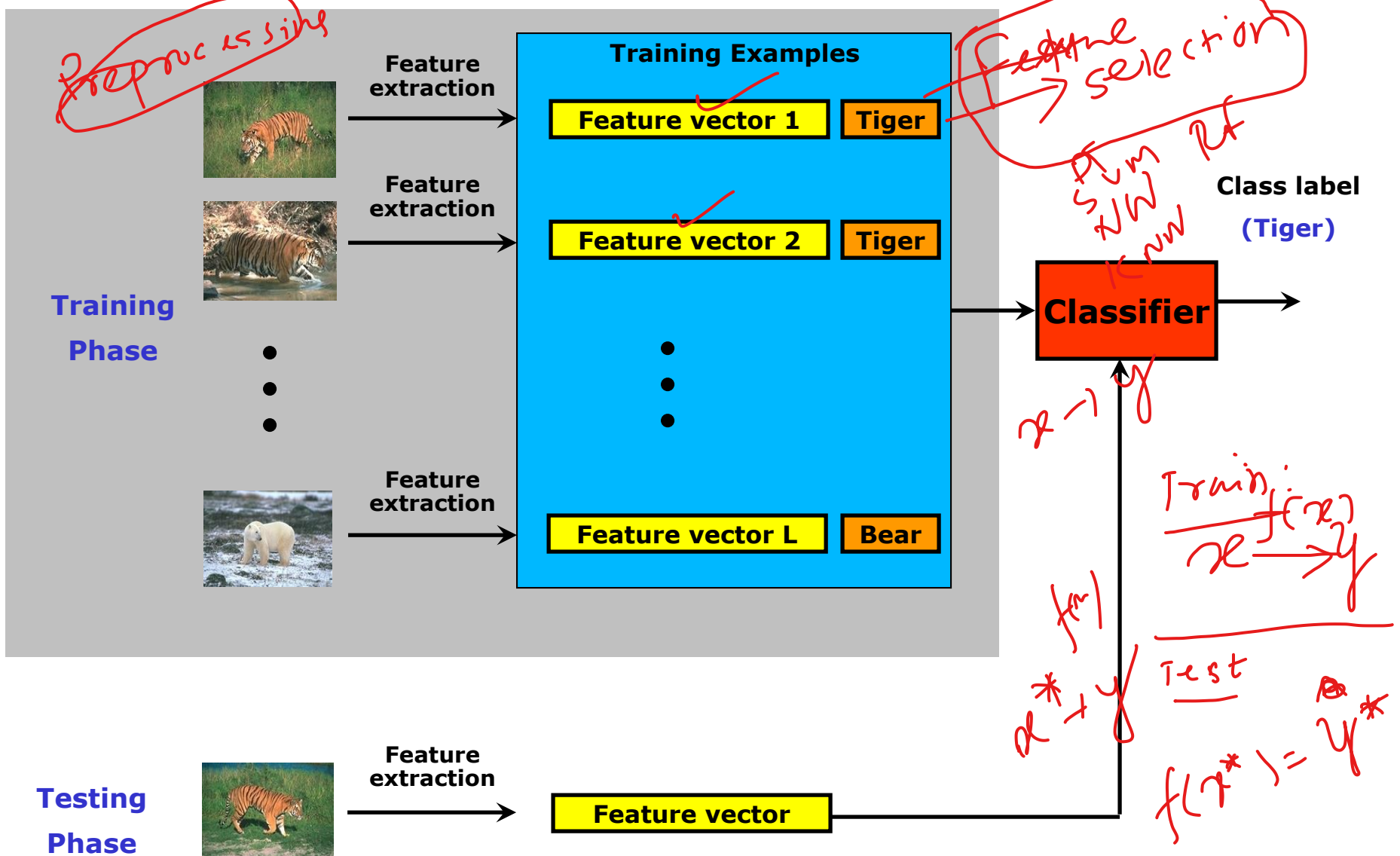
$x_2 = [\quad \quad \quad]$

- ❑ We assume that each pattern is represented **uniquely** by a single feature vector and that it can belong to only one class.

$x_3 = [\quad \quad \quad]$

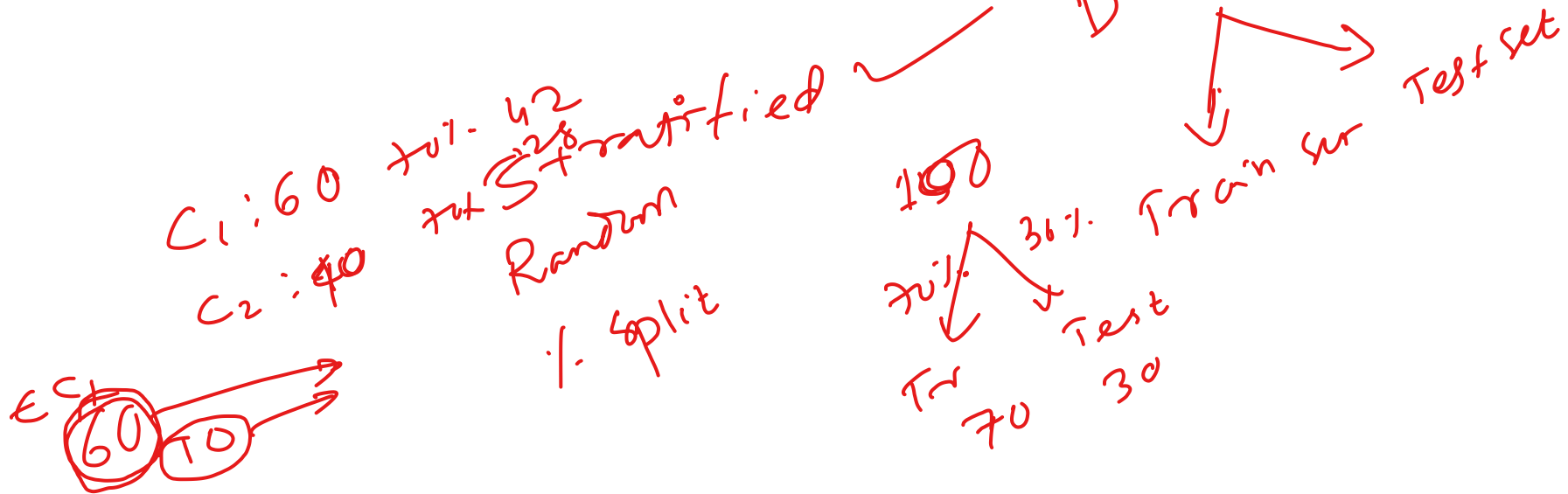
Pattern Classification: Example

Static pattern: An example is represented by a vector of features



Evaluation Measures

- ❑ Success: actual output = target
- ❑ Error: actual output $\neq$ target
- ❑ % error = $\frac{\text{\#errors}}{\text{\#samples}}$



Evaluation Measures

- ❑ Training: In training the model is built
- ❑ Testing: In testing, the model is applied to the new data.

The goal in building a machine learning algorithm is to perform well on both training and testing data.

- ❑ Error on training data is called training error.
- ❑ Error on test data is called test error.
- ❑ Test error indicates how well model will perform on new data. (**Generalization**)

Performs well on new data $\Rightarrow$ Good generalization
Test error = Generalization error

Overfitting and Underfitting

Tr - 95%
Test - 50%

- ❑ **Overfitting:** If a model has **low training error** and **high test/generalization error**, then it is called overfitting
- ❑ Fit to the noise in the training data

Overfitting = Poor generalization

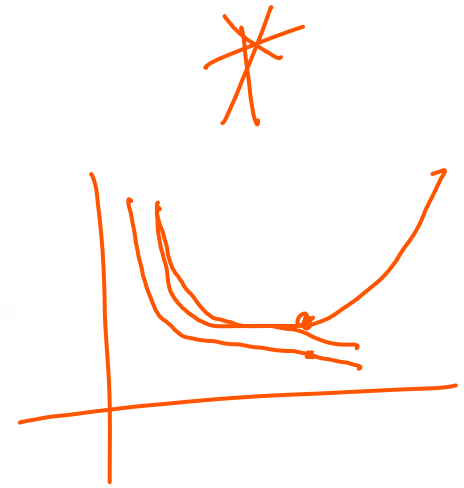
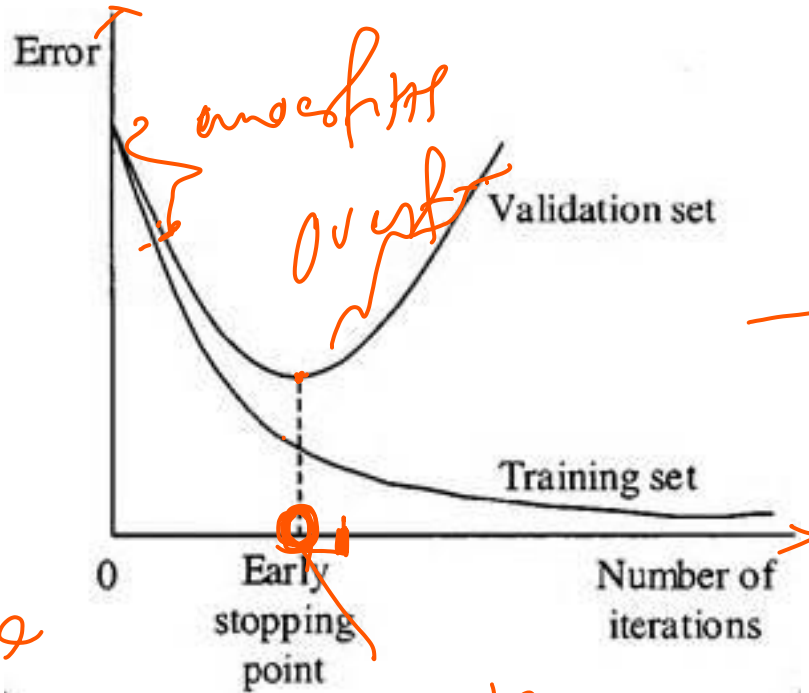
✓
Complex model

- ❑ **Underfitting:** **high training error** and **high test/generalization error**
- ❑ **The cause of the poor performance of a model in machine learning is either overfitting or underfitting the data.**

How to prevent Overfitting?

→ Early Stopping
Validation set

D
↓
Train
↓
Test



Train validate

Perf. Measures / Metrics W & B
Confusion matrix
- Classifier
- Regression model
- Clustering
- Unsupervised

K-NN

$K = 3, 5, 7, 9, 11$
 $K = ?$
valid
AIC

80/100

Confusion matrix

test set
40 40
C1 C2
+ve -ve
Ans

Confusion Matrix is a tool to determine the performance of classifier. It contains information about actual and predicted classifications.

2x2
dummy classification
+ve -ve
TP TN
FP FN
Actual
40
60

Predicted

	Positive	Negative
Positive	35 TP	0 FN
Negative	10 FP	50 TN

85
Acc = $\frac{100}{TP+TN}$
40 = $\frac{TP+TN}{TP+FP+FN+TN}$
60 TPR or Sen
Recall = $\frac{TP}{TP+FN}$

True Positive (TP): # positive samples correctly identified as positive.

False Negative (FN): # positive samples incorrectly identified as negative

False Positive (FP): # negative samples incorrectly identified as positive

True Negative (TN): # negative samples correctly identified as negative

1 sec
TNR = $\frac{50}{60} = \frac{TN}{TN+FP}$

Accuracy, sensitivity, specificity

Data
imbalanced

sensitivity = true positive rate
= recall
= 1 - false negative rate
= $TP / (TP + FN)$

specificity = true negative rate
= 1 - false positive rate
= $TN / (TN + FP)$ ✓

✓ accuracy = $(TP + TN) / (TP + TN + FP + FN)$

✓ precision = $TP / (TP + FP)$

~~*~~ F1 Score = $2 \times \frac{Precision \times Recall}{Precision + Recall}$

Bayesian / w
from spec /
= $\frac{sen + spe}{2}$ 10 90
cr
G
+ve -ve
pred
true 10 FN
0 FN 90 TN

Acc = 90%

2 on terms of
TP, TN, FP, FN

Confusion matrix

3, 4, 5, 6 - - -
↓
C₁ C₂ C₃

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Regression Error Metrics

Mean Absolute Error: average of the difference between the Original Values and the Predicted Values. It gives us the measure of how far the predictions were from the actual output.

The diagram illustrates the Mean Absolute Error (MAE) formula with the following components and annotations:

- Divide by the total number of data points:** Points to the $\frac{1}{n}$ term.
- Sum of:** Points to the summation symbol $\sum$.
- Actual output value:** Points to the y term inside the absolute value.
- Predicted output value:** Points to the $\hat{y}$ term inside the absolute value.
- The absolute value of the residual:** Points to the entire absolute value expression $|y - \hat{y}|$.

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

$$MAE = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|$$

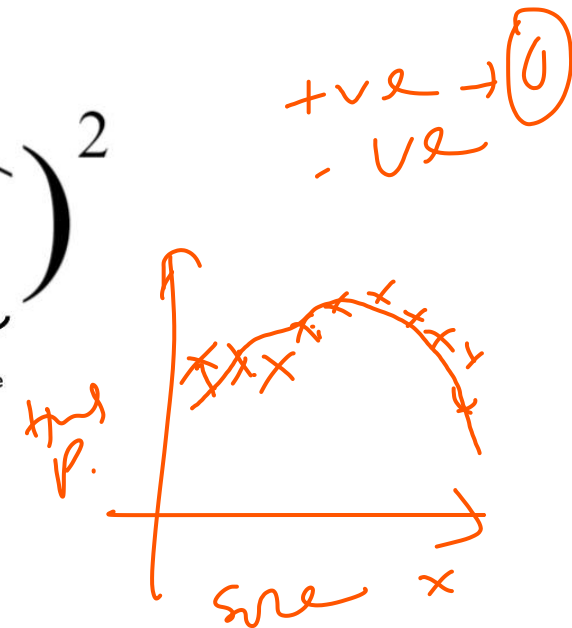
Regression Error Metrics

Mean Square Error (MSE): average of the **square** of the difference between the original values and the predicted values. The advantage of MSE being that it is easier to compute the gradient, whereas Mean Absolute Error requires complicated linear programming tools to compute the gradient

$$MSE = \frac{1}{n} \sum_i \left(\underbrace{y^t - \hat{y}}_{\substack{\text{The square of the difference} \\ \text{between actual and} \\ \text{predicted}}} \right)^2$$

Root Mean Square Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



Regression Error Metrics

Mean absolute percentage error (MAPE):

$$MAPE = \frac{100\%}{n} \sum \left| \frac{\overbrace{y - \hat{y}}^{\text{The residual}}}{\underbrace{y}_{\text{Each residual is scaled against the actual value}}} \right|$$

Multiplying by 100% converts to percentage

Mean percentage error:

$$MPE = \frac{100\%}{n} \sum \left(\frac{y - \hat{y}}{y} \right)$$